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Simulation study on the performance of low-temperature water gas shift membrane reactor system

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HIGHLIGHTS

- A 2D symmetric model of WGSR membrane reactor is built.
- Effect of dimensionless parameters and temperature is analyzed.
- Effect of sweep gas flow rates and sweep gas modes is investigated.
- Two indicators are proposed to guide the design of reactor geometric length.

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ABSTRACT

In order to understand the behavior of the low-temperature water gas shift (WGS) reaction membrane reactor system and obtain better performance of the reactor, numerical simulations are performed with a two-dimensional axisymmetric model. Dimensionless analysis is applied to the governing equations, and then a set of dimensionless parameters are established. The effect of the parameters of Damkohler number (*Da*) and membrane permeability (*Pp*) over a wide range on the performance of CO conversion and hydrogen recovery is studied, and the discussion of reaction phenomenon is presented. The WGS reaction is simulated under the operating conditions of 2 atm and two temperatures of 200 and 300 °C. The simulation results suggest that $Da = 10^{-4}$ and $Pp = 10^{-1}$ are the optimum operating conditions for the reactor system where the conversion reaches nearly 100% exceeding the thermodynamic equilibrium conversion by 11% at 300 °C and the optimal hydrogen recovery is achieved simultaneously. The distribution of components along the axis of the reactor is checked, and then two crucial indicators are raised to assess the extent of reaction and hydrogen separation respectively, for the purpose of guidance for the length design of the reactor. Finally, the analysis of the effect of sweep gas on reactor performance suggests that the optimum flow rate of sweep gas is 0.2–0.5 L/min, and further enhancement of performance can be achieved by using counter-current sweep gas configuration.

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Introduction

The water gas-shift (WGS) reaction has a large number of applications in the hydrogen production and the ammonia synthesis industry using coal, oil and natural gas as raw materials. It is also commonly used for carbon monoxide removal in hydrogen production via steam reforming and new technologies such as hydrogen fuel cells (HFCs) [1] and integrated gasification combined cycle (IGCC) [2–5].

$$CO + H_2O \Leftrightarrow H_2 + CO_2 \quad \Delta H = -41.1 kJ/mol \tag{1}$$

Due to its exothermic nature, it is difficult to achieve a high conversion at a fast reaction rate under high temperature. As a result, the WGS reaction process is implemented in two steps in the industry: rapid reaction at high temperature (>350 °C), and enhancement of reaction extent at low temperature (<300 °C) [6]. According to the difference of WGS temperature, the catalysts can be roughly divided into two types: high-temperature shift catalysts and low-temperature shift catalysts. The former are mainly based on Fe–Cr and Ni, while the latter are mainly based on Cu [7,8]. In addition, some noble metal catalysts such as Pt have shown excellent catalytic performance compared with traditional catalysts, making it possible to be the next generation WGS reaction catalysts [9].

Compared with the traditional fixed-bed reactors, the catalytic membrane reactors can simultaneously perform WGS reaction and hydrogen separation, thereby obtaining higher reaction conversion and hydrogen purity. To date, there has been extensive research on membrane reactors, which is not limited to WGS reaction, but also includes various steam reforming reactions [3,10–12]. Hydrogen-selective membranes are the key to achieve high performance of membrane reactors, because removing hydrogen can change the composition of the reaction zone and thus break the chemical equilibrium in line with Le Chatelier's principle [13]. At present, there are many kinds of membranes such as Pd-based, MFI zeolite-based, silica and CMS membranes [14–17], among which Pd-based membranes are considered to be ideal membranes for hydrogen separation and purity due to their high selectivity and permeability, and have been widely studied in catalytic membrane reactors [7]. Tosti et al. [18] conducted a long-term test on the performance of the Pd–Ag alloy membrane over one year. The results showed that the membrane hydrogen permeability was stable around at 1.0×10^{-8}mol $m^{-1}s^{-1}Pa^{-0.5}$ measured at 350–400 °C, and no defect or crack was found in the membrane. This implied that the Pd-based membrane not only had extremely high permeability, but also had good stability and durability.

In the past two decades, a considerable amount of literature has reported the performance and feasibility of WGS reaction catalytic membrane reactors from different aspects. Chein et al. [19] established a two dimensional steady-state model of high temperature WGS reaction, mainly studying the influence of membrane permeability and steam to carbon ratio on the carbon monoxide conversion. Their results suggested that at low permeability the CO conversion was close to that of traditional reactors, while at high permeability the CO conversion was greatly improved. Based on the experimental results, Hla et al. [20] established a two dimensional non-isothermal model and used it to analyze the effect of catalyst content and temperature on the reactor performance. They found that increasing the amount of catalyst and reaction temperature could achieve high reactor performance, but also pointed out that excessive catalyst content and inlet temperature would induce an increase in temperature, leading to a deviation from the reasonable temperature. Adrover et al. [21] mainly discussed the CO conversion and thermal effect of WGS reaction varying with operating pressure. They claimed that the conversion increased obviously with increasing operating pressure and the temperature inside the reactor would also rise slightly.

In fact, there are many other factors affecting membrane reactor performance including, space velocity, material diffusion, reaction kinetics and reactor geometry in addition to the factors mentioned above, and the interplay among them can-not be ignored. However, most published papers do not pay attention to the interaction between multiple factors, reactor configuration and operating conditions varying in different work, which makes some conclusions do not have universal applicability. Brunetti et al. [22,23] first proposed the volume index (VI) to analyze the advantage of membrane reactors over fixed-bed reactors, and then studied the effect of dimensionless parameter Damkohler number on reactor performance. Murmura et al. [24,25] applied Darcy's law to the gas flow through porous media in membrane reactors and established several crucial dimensionless parameters. By discussing the dimensionless parameters, the factors significantly affecting the performance of the membrane reactor under different operating conditions were determined. Besides, WGS reaction in catalytic membrane reactors is generally carried out under high temperature conditions, and less research is conducted under low temperatures.

Herein, in this work a two-dimensional axisymmetric physical model of membrane reactor was established, based on low-temperature WGS reaction kinetics, and then the governing equations of the model were directly dimensionless to obtain dimensionless parameters. The aim here is to quantitatively analyze the effects of fluid flow rate, reaction rate, and permeate rate on the membrane reactor performance, providing some guidance for the design of membrane reactors. The main innovation of this paper is to use dimensionless parameters to analyze the membrane reactor system. And how the WGS reaction kinetics determine reaction behavior inside the reactor is discussed in detail. In addition, in order to ensure the accuracy of the model, the numerical results are compared with the experimental results.

Physical and mathematical models

Membrane reactor

The physical model of the membrane reactor is shown in Fig. 1(a). Generally, a tubular membrane reactor is consisted of an internal tube, a shell and a membrane. Low temperature catalyst is placed in the inner tube forming the reaction zone, while the annular area becomes the permeation zone. This configuration is very common in the studies of membrane reactors [19,26–28]. A certain proportion of water vapor,

carbon monoxide and other gas mixtures undergo chemical reaction in the reaction zone producing carbon dioxide and hydrogen. Due to the presence of the sweep gas in the permeation zone, the hydrogen partial pressure in the permeation zone is lower than that in the reaction zone, and the product hydrogen continuously flow from the retentate side to the permeate side through the membrane. Hydrogen flux permeating through the membrane is usually defined by two variables, pressure and temperature:

$$F_{H_2} = P_m\left(p_1^n - p_2^n\right), P_m = \frac{P_L}{\delta}\exp\left(-\frac{E_H}{RT}\right) \quad (2)$$

where δ is the membrane thickness, P_L is the membrane permeability constant, E_H is the membrane activation energy, P_m is the membrane permeance, p_1 and p_2 are the hydrogen partial pressures on the retentate and permeate sides, respectively. In particular, when n is equal to 0.5, Eq. (2) degenerates into Sieverts' law which is widely adopted in membrane reactors research.

The carbon monoxide reacts rapidly with water at the reactor entrance, and the concentration gradient of reactant changes sharply. Therefore, it is necessary to extend proper geometric length at the reactor inlet in this model to simulate accurately the membrane reactor. Besides, assuming that the catalyst is evenly distributed in the reaction zone, the physical model can be simplified to two-dimensional axisymmetric, as is shown in Fig. 1(b).

WGS reaction kinetics

Generally, the micro mechanism of WGS kinetics can be divided into two types: associative mechanism and regenerative mechanism. Up to now, based on two kinds of mechanisms, the study of WGS reaction kinetics can be found everywhere. The common low-temperature water gas reaction rate expressions are listed in Table 1.

As can be seen from Table 1, whether it is a kinetic expression derived from microscopic reaction mechanisms or an empirical rate expression derived from experiments, the reaction rate can be written in the following general form:

$$r = k \cdot A\left(P_{CO}, P_{H_2}, P_{CO_2}, P_{H_2}\right) \cdot (1-\beta) \quad (3)$$

$$\beta = \frac{P_{CO_2}P_{H_2}}{P_{CO}P_{H_2O}}\frac{1}{K_e} \quad (4)$$

where k is the reaction rate constant, K_e is the thermodynamic equilibrium constant, P_i is the partial pressure of component i in the reaction zone. Considering the rate expression derived by Campbell, A is the following equation:

$$A\left(P_{CO}, P_{H_2O}, P_{CO_2}, P_{H_2}\right) = \frac{P_{CO}P_{H_2O}}{\left(1 + K_{CO}P_{CO} + K_{H_2O}P_{H_2O} + K_{CO_2}P_{CO_2} + K_{H_2}P_{H_2}\right)^2} \quad (5)$$

where K_i is adsorption constant for component i.

Naturally, the rate expression can be regarded as composed of three parts which are k, A and $1-\beta$. $1-\beta$ quantifies the distance of the reaction from chemical equilibrium. A is closely related to species fraction and adsorption of species. And both of them can be seen as functions of temperature. Thus, the reaction process inside the reactor can be clarified by analyzing k (replaced by Da mentioned later) and temperature. In this way, the complex reaction kinetics is greatly simplified.

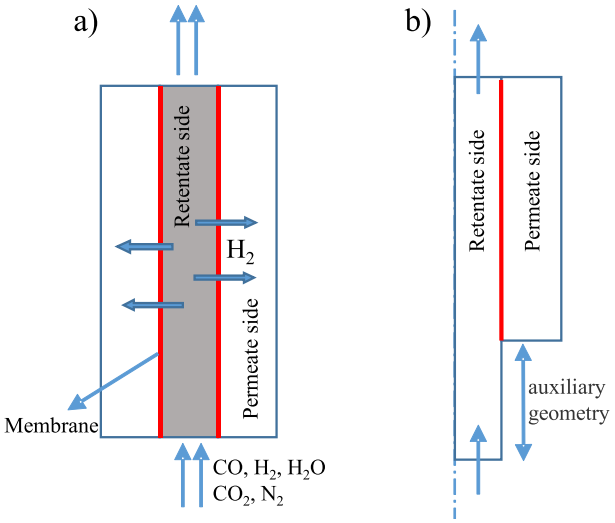


Fig. 1 – Geometric model of membrane reactor.

Mathematic development of the model

Before building the mathematical equations, according to the geometry of the membrane reactor and the operating conditions where the reaction takes place, the following assumptions are adopted to simplify the reactor model:

a) Steady state and isothermal reaction (mild exothermicity)
b) Ideal gas behavior and laminar flow (Reynolds number = 5)
c) Regardless of the permeable membrane thickness
d) Only hydrogen passing through the membrane (infinite selectivity of Pd-based membrane)

Governing equations

Mass continuity equation:

for porous media:

$$\nabla \cdot (\rho \mathbf{u}) = P_m\left(p_1^n - p_2^n\right)M_H \quad (6)$$

for permeate side and fluid entrance region:

$$\nabla \cdot (\rho \mathbf{u}) = 0 \quad (7)$$

Table 1 – Low-Temperature WGS reaction kinetic models [29].

ref	expressions	mechanisms
Campbell	$\frac{kP_{CO}P_{H_2O}(1-\beta)}{(1+K_{CO}P_{CO}+K_{H_2O}P_{H_2O}+K_{CO_2}P_{CO_2}+K_{H_2}P_{H_2})^2}$	associative mechanism
Shchibrya	$\frac{kP_{CO}P_{H_2O}(1-\beta)}{AP_{H_2O}+P_{CO_2}}$	redox mechanism
Temkin	$kP_{CO}\left(\frac{P_{H_2O}}{P_{H_2}}\right)^{0.5}(1-\beta)$	redox mechanism
Moe	$kP_{CO}P_{H_2O}(1-\beta)$	empirical expression
Goodgidge	$kP_{CO}^{a}P_{H_2O}^{b}P_{CO_2}^{c}P_{H_2}^{d}(1-\beta)$	empirical expression

where ρ is gas mixture density, $\boldsymbol{u}$ is the flow velocity, M_H is the molar mass of hydrogen.

Momentum equation

for porous media (Darcy's law):

$$\boldsymbol{u}=-\frac{\kappa}{\mu}\nabla p \tag{8}$$

for permeate side and fluid entrance region:

$$\rho(\boldsymbol{u}\cdot\nabla)\boldsymbol{u}=\nabla\cdot\left[-pI+\mu\left(\nabla\boldsymbol{u}+(\nabla\boldsymbol{u})^{\top}\right)-\frac{2}{3}\mu(\nabla\cdot\boldsymbol{u})I\right] \tag{9}$$

where μ is the viscosity, κ is the packed bed permeability, p is the pressure.

Species balance equations based on Maxwell-Stefan diffusion model

$$\nabla\cdot\boldsymbol{J}_i+\rho(\boldsymbol{u}\cdot\nabla)\omega_i=R_i \tag{10}$$

$$R_i=kA(1-\beta)M_i \tag{11}$$

$$\boldsymbol{J}_i=-\left(\rho\omega_i\sum_k D_{i,k}\boldsymbol{d}_k\right) \tag{12}$$

$$\boldsymbol{d}_k=\nabla x_k+\frac{\nabla p}{p}(x_k-\omega_k) \tag{13}$$

where $\boldsymbol{J}_i$, R_i respectively denotes the mass change of component i caused by diffusion and chemical reaction, $D_{i,k}$ is the multicomponent Fick diffusion coefficient, x_k, ω_k are the mole fraction and mass fraction of component k, respectively, k is the reaction rate, M_i is the molar mas of component i.

Hydrogen flux at the membrane:

$$-\boldsymbol{n}\cdot(\boldsymbol{J}_i+\rho w_i\boldsymbol{u})=P_m\left(p_1^n-p_2^n\right)M_H \tag{14}$$

generally, when Sieverts' law is used to calculate the flux through the membrane, n is equal to 0.5.

Boundary conditions

Laminar flow at tube side (dependent variables $\boldsymbol{u}_1$, p_1)

$$\text{inlet}: \boldsymbol{u}_1=-u_0\boldsymbol{n} \tag{15}$$

$$\text{outlet}: p_1=-p_2\boldsymbol{n} \tag{16}$$

Reacting flow at tube side (dependent variables $\boldsymbol{u}_2$, p_2)

$$\text{inlet}: -\boldsymbol{n}\cdot\rho\boldsymbol{u}_2=\rho|\boldsymbol{u}_1| \tag{17}$$

$$\text{outlet}: p_2=P_{tube} \tag{18}$$

$$\text{membrane}: -\boldsymbol{n}\cdot\rho\boldsymbol{u}_2=-F_{H_2} \tag{19}$$

Species at tube side (dependent variables ω_i)

$$\text{inlet}: \omega_i=\omega_{in,j},\ \omega_{in,j}: -\int_{\partial\Omega}\boldsymbol{n}\cdot\left(\boldsymbol{j}_i+\rho\boldsymbol{u}_1\omega_{in,j}\right)dS=Q_{in,j} \tag{20}$$

$$\text{outlet}: -\boldsymbol{n}\cdot\rho\omega_i\sum_k D_{i,k}\boldsymbol{d}_k=0 \tag{21}$$

$$\text{membrane}: -\boldsymbol{n}\cdot\left(\boldsymbol{j}_i+\rho\boldsymbol{u}_2\omega_i\right)=F_i(i=H_2) \tag{22}$$

Shell side (dependent variables $\boldsymbol{u}_3$, p_3, ω_i)

$$\text{inlet}: \boldsymbol{u}_3=-u_{sweep}\boldsymbol{n},\ \omega_i=\omega_{in,j} \tag{23}$$

$$\text{outlet}: p_3=-P_{shell},\ -\boldsymbol{n}\cdot\rho\nabla\omega_i=0 \tag{24}$$

$$\text{membrane}: -\boldsymbol{n}\cdot\boldsymbol{j}_i=F_i(i=H_2) \tag{25}$$

Nondimensionalization

Defining characteristic physical quantity:

$$x'=\frac{x}{d},y'=\frac{y}{d},z'=\frac{z}{d},\nabla'=d\nabla \tag{26}$$

$$\boldsymbol{u}'=\frac{\boldsymbol{u}}{u_0},\rho'=\frac{\rho}{\rho_0},D'_{i,k}=\frac{D_{i,k}}{D_0},p'=\frac{p}{P_0},T'=\frac{T}{T_0},M'_i=\frac{M_i}{M_0} \tag{27}$$

where d is the reactor diameter, u_0 is the inlet velocity, ρ_0 and M_0 are respectively the gas mixture density and the average molar mass at the reactor entrance, T_0 and P_0 are reaction temperature and pressure, D_0 is the diffusion coefficient, which is assumed to be $1e-5m^2/s$.

The dimensionless forms of Eqs. (10)–(14) are as follow:

$$\boldsymbol{d}_k=\frac{1}{d}\left(\nabla' x_k+\frac{\nabla' p'}{p'}(x_k-\omega_k)\right)=\frac{1}{d}\boldsymbol{d}'_k \tag{28}$$

$$\boldsymbol{J}_i=-\left(\frac{\rho_0 D_0}{d}\right)\left(\rho'\omega_i\sum_k D'_{i,k}\boldsymbol{d}'_k\right)=\left(\frac{\rho_0 D_0}{d}\right)\boldsymbol{J}'_i \tag{29}$$

$$\left(\frac{\rho_0 D_0}{d^2}\right)\nabla' \cdot J_i' + \left(\frac{\rho_0 u_0}{d}\right)\rho' \boldsymbol{u}' \cdot \nabla' \omega_i = (kAM_0)(1-\beta)M_i' \tag{30}$$

$$-\boldsymbol{n} \cdot \left(\frac{D_0}{u_0 d} J_i' + \rho' w_i \boldsymbol{u}'\right) = \frac{P_m P_0^n (p_1'^n - p_2'^n) M_H}{\rho_0 u_0} \tag{31}$$

Applying the ideal gas law $P_0 = \frac{\rho_0 R T_0}{M_0}$, Eqs. (30) and (31) can be rewritten as:

$$\left(\frac{D_0}{u_0 d}\right) J_i' + \rho' \boldsymbol{u}' \cdot \nabla' \omega_i = \left(\frac{R T_0 k d}{P_0 u_0}\right) A(1-\beta) M_i' \tag{32}$$

$$-\boldsymbol{n} \cdot \left(\frac{D_0}{u_0 d} J_i' + \rho' w_i \boldsymbol{u}'\right) = \frac{R T_0 P_m P_0^{n-1} (p_1'^n - p_2'^n) M_H}{u_0 M_0} \tag{33}$$

Thus, the dimensionless parameter group can be obtained, which are:

$$Da = \frac{R T_0 k d}{P_0 u_0}, \frac{1}{Pe} = \frac{D_0}{u_0 d}, Pp = \frac{R T_0 P_m P_0^{n-1}}{u_0} \tag{34}$$

Eqs. (32) and (33) can be rewritten as:

$$\left(\frac{1}{Pe}\right) J_i' + \rho' \boldsymbol{u}' \cdot \nabla' \omega_i = (Da) A(1-\beta) M_i' \tag{35}$$

$$-\boldsymbol{n} \cdot \left(\frac{1}{Pe} J_i' + \rho' w_i \boldsymbol{u}'\right) = \frac{Pp (p_1'^n - p_2'^n) M_H}{M_0} \tag{36}$$

where the dimensionless parameter *Da* (Damkohler number) is the ratio of the chemical reaction rate to the convection rate, The dimensionless parameter *Pp* representing the membrane permeability is the ratio of the hydrogen separation rate to the convection rate, and *Pe* (Peclet number) representing the diffusivity is the ratio of the convection rate to the diffusion rate. It is worth noting that the dimensionless parameters are all related to the convection rate, which means that all the above parameters will change simultaneously when the inlet velocity changes. Therefore, in order to study the effect of each parameter on the reactor performance independently, the reactor gas inlet velocity is kept constant at a rational value. In addition, it should be pointed out that among multiple kinetic parameters *k* is the only one that appears in the dimensionless parameter *Da*, while the other kinetic parameters (*A* and $1-\beta$) as functions of temperature are not included. This implies that variations of reaction field inside the membrane reactor are inevitable at different temperatures in spite of the same *Da*. Hence it is at different temperatures that it is necessary to study the behaviors of membrane reactors. For this reason, the following study in this article is conducted at high and low temperatures (473 K and 573 K).

Generally, the CO conversion and hydrogen recovery are the key index for evaluating the performance of WGS membrane reactors, which are defined as follows:

$$X_{CO} = \frac{CO_{in} - CO_{out}}{CO_{in}} \tag{37}$$

$$R_1 = \frac{H_p}{H_p + H_{out}} \tag{38}$$

$$R_2 = \frac{H_p}{H_{in}} \tag{39}$$

where CO_{in} , CO_{out} respectively represent the CO mass flow at the inlet and outlet of the reactor, H_p , H_{out}, H_{in} respectively represent the hydrogen mass flow passing through the membrane, flowing out at the reactor outlet and flowing into the reactor, X_{CO} is the carbon monoxide conversion. R_1 is the hydrogen recovery, which is used to evaluating the overall hydrogen separation efficiency of the membrane reactor in this work, indicating the separation degree of hydrogen produced by chemical reaction when there is no hydrogen inflow at the reactor inlet. R_2 is the ratio of the total permeated hydrogen to the hydrogen flowing into the reactor.

Numerical method

To quantitatively analyze the influence of each dimensionless variable on the performance of the membrane reactor, the finite element software COMSOL is used to solve the Eqs. (6)–(14) to simulate the fluid flow and chemical reaction in the reactor. The mesh is refined in the area at the junction of flow and reaction and the area next to the membrane, and grid independence verification is also carried out, which are not shown in this article for the sake of brevity.

Results and discussion

The research of this paper is carried out based on the operating conditions of the study of Mendes et al. [30] which are listed in Table 2. The parameters of WGS reaction kinetics are adopted from the study of Mendes et al. [31]. The feed gas is composed of 4.7% CO, 34.8% H_2O, 28.7% H_2, 10.2% CO_2 and diluent N_2, whose composition is derived from ethanol autothermal reforming products [32], while a high speed N_2 flow is employed on the permeate side to minimize the effect of hydrogen accumulation.

The comparison of CO conversion between the calculated results and experimental results is shown in Fig. 2. Except for slight difference between the two sets of data when CO

Table 2 – Operating conditions.

Parameter name	Value
Operating temperature T	473–573 K
Operating pressure at retentate side P_0	2atm
Operating pressure at permeate side P_1	1atm
Reactor length z	0.05 m
Reactant side radius r_1	0.005 m
Permeate side radius r_2	0.0175 m
Membrane thick δ	50 μm
Membrane permeance Pm	$5.45 \times 10^{-8} m^{-1} s^{-1} Pa^{-0.5}$
Membrane activation energy Em	10.72 kJ/mol
Catalyst content Wcat	1.5 g
GSHV	1200-10800L_N kg^{-1}_{cat} h^{-1}
Porosity ε	0.4
N2 sweep gas flow	1 L/min

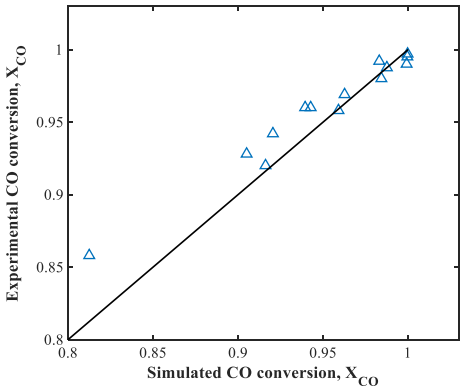


Fig. 2 – Parity plots of calculated and experimental results for H_2 conversion.

conversion is at a low value, it can be seen clearly that the calculated results of the model are in good agreement with the experimental results. And the difference is also consistence with that reported in the literature [30] for the same reason.

Then, based on experiments and some literatures, the research variables *Da* and *Pp* are set to span two or even three orders of magnitudes to ensure the applicability of simulation and accuracy of this simulation, while P_e is always maintained at about 5 because of the constant inlet velocity. Table 3 lists the approximate values or range of the dimensionless parameters calculated according to the definitions in this article.

Overall performance of the membrane reactor

Since the Damkohler number is mentioned in a considerable research on the WGS reaction membrane reactor, this article also begins with the analysis of the effect of *Da* number on the carbon monoxide conversion. Fig. 3 mainly presents the effect of *Da* number on the CO conversion when the WGS reaction takes place at a relative high temperature of 573 K and a low temperature of 473 K. And the thermodynamic equilibrium conversion curves are also added to the figure to clearly illustrate the role played by membrane in separating hydrogen. Fig. 3(a) and (b) both depict the same fact that whether the reaction is conducted under high or low temperature, the conversion increase with the increase of *Da*, gradually approaching to be constant. But evidently, under different membrane permeability the reasons why the conversion no longer increases are different. When membrane permeability $P_p = 10^{-3}$ (ultra-low permeability), it is taken for granted that almost no hydrogen permeates through the membrane, and the conversion is limited by thermodynamic equilibrium as illustrated in Fig. 3(a). In contrast, when $Pp = 10^{-1}$ (high membrane permeability) the conversion almost reaches 100%, indicating that carbon monoxide is completely consumed. In fact, when *Da* is equal to 10^{-4}, Xco is already very close to the maximum value as shown in Fig. 3. In the light with the formula of *Da*, when achieving the same conversion, the lower *Da* value means the lower catalyst content required for the reaction or the higher the corresponding gas flow velocity, which undoubtedly benefits the

Table 3 – Values of dimensionless parameters.

Temperature(°C)	*Da*	*Pp*	*Pe*	Membrane	Thick	Refs
200 – 300	$3.5 \times 10^{-5} - 3.16 \times 10^{-4}$	0.1	5.5 – 6.6	Pd–Ag	50 μm	[30]
400 – 700	–	0.82 – 2	64.8 – 94	Pd-based	5 μm	[33]
150 – 300	–	0.012 – 0.0027	5.9	zeolite		[34]
250 – 350	–	–	75			[35]
200/300	$10^{-5} - 10^{-3}$	$10^{-3} - 10^{-1}$	5	Pd-based		This work

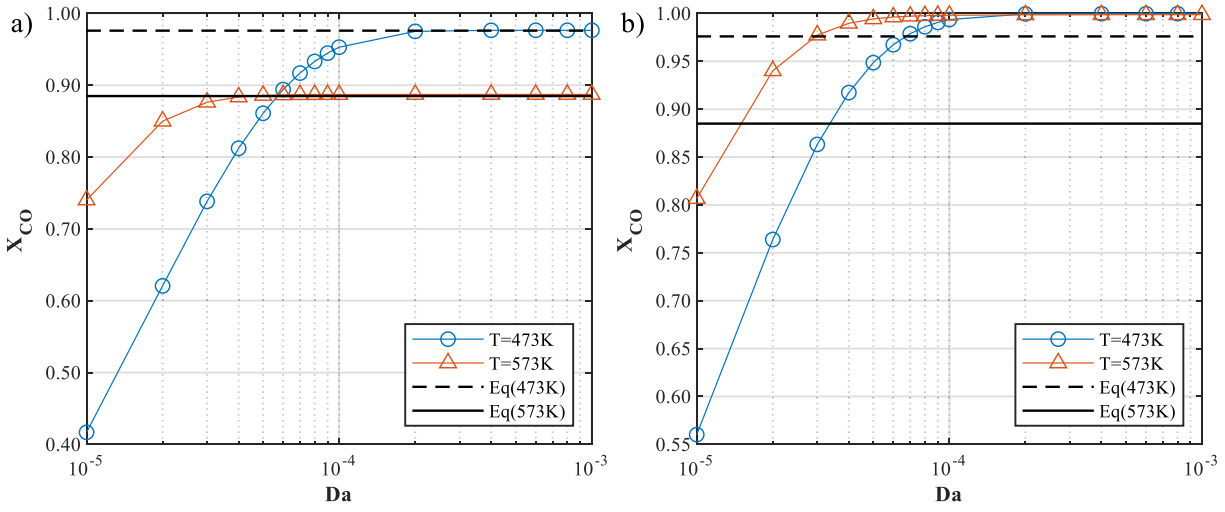


Fig. 3 – Effect of *Da* and T on CO conversion at (a) $Pp = 10^{-3}$, (b) $Pp = 10^{-1}$. The black dotted lines and solid lines symbol the thermodynamic equilibrium conversion at T = 473 K and 573 K, respectively.

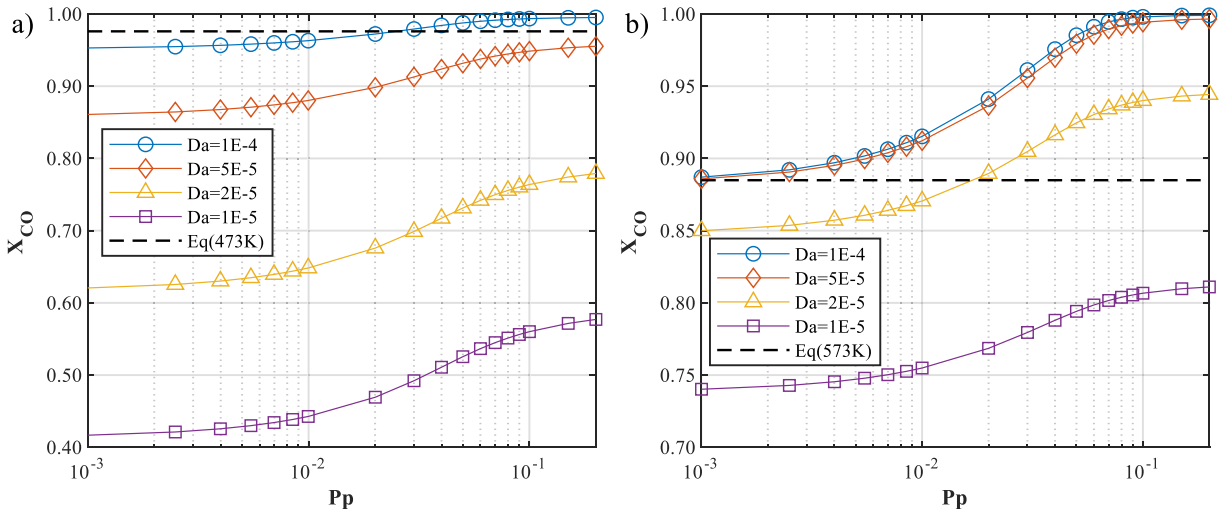


Fig. 4 – Effect of *Pp* and T on CO conversion at(a) T = 473 K, (b) T = 573 K. The black dotted lines symbol the thermodynamic equilibrium conversion at (a) T = 473 K, (b) T = 573 K.

design and cost reduction of membrane reactors. Besides, the comparison of CO conversion at two temperatures of 573 K and 473 K in Fig. 3(a) and (b) suggests that enhancement in conversion attributed to the effect of hydrogen separation is remarkable at high temperatures but that is not apparent at low temperatures.

At two different temperatures, the carbon monoxide conversion under different membrane permeability is shown in Fig. 4. The examined variable *Pp* is varied from 10^{-3} to 10^{-1}, and *Da* is set in the range of $10^{-5} - 10^{-4}$. From the logarithmic graph, Xco increases significantly as *Pp* increases from 10^{-2} to 10^{-1}. Further increase of *Pp* results in only a small enhancement of Xco. For $Pp < 10^{-3}$, the membrane reactor can be regarded as the traditional fixed-reactor, although Xco exceeds the equilibrium conversion in some cases as shown in Fig. 4(b), which is also reported in the literature [19] for the reason that *Pp* is not exactly equal to zero. Especially for $Pp = 10^{-1}$ and $Da = 10^{-4}$, Xco approaches the limited value of 100%. Nevertheless, it is worth noting that for $Pp \geq 10^{-1}$ and $Da = 10^{-5}$ Xco is still lower than the equilibrium conversion despite not being restricted by thermodynamics. This can be explained well in Fig. 5, which shows the CO and hydrogen

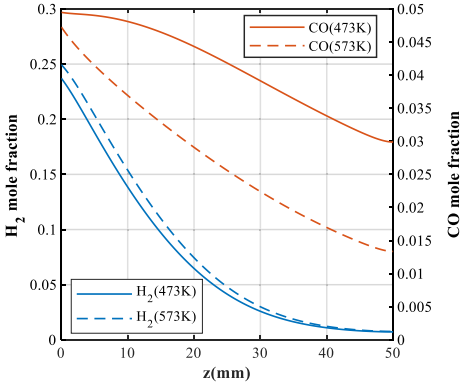


Fig. 5 – CO and H_2 mole fraction along the reactor at $Da = 10^{-5}$, $Pp = 10^{-1}$.

mole fraction along the axial direction of the reactor (along reactor centerline). Here at two temperatures, the CO fractions are different; instead both the hydrogen fractions are almost the same. The hydrogen fractions are nearly equal to zero at the outlet of the reactor, while the proportion of unreacted CO still accounts for about 63% (473 K) and 28% (573 K), respectively. Hence the result illustrated in Fig. 5 indicates that the rate of hydrogen separation is greater than that of hydrogen production through WGS reaction under this condition. In other words, the performance of membrane reactors is mainly restricted by chemical reaction in the case of large membrane permeability and low *Da* values.

As another important index of membrane reactor performance, hydrogen recovery R_1 is often adopted to evaluate the degree of hydrogen separation by the membrane. Ideally, $R_1 = 1$ means that there is no hydrogen at the outlet of the retentate side. The effect of *Pp* on R_1 is shown in Fig. 6(a). As expected, R_1 increases with the increase of *Pp* and it approaches to unity for $Pp = 10^{-1}$ that means hydrogen in the reaction zone is completely separated. Unfortunately, the curves of R_1 at different *Da* values are almost coincided because a large account of hydrogen is already present in the reactor before WGS reaction happens. A similar observation is also reported in the literature [35] for the same reason, demonstrating that the single R_1 cannot precisely explain the effect of chemical reaction rate on the reactor performance when the feed gas contains hydrogen.

Therefore, another hydrogen recovery index R_2 that has no limit theoretically is adopted to evaluate the effect of reaction enhanced by deviations from thermodynamically equilibrium. The smaller the R_2 is, the worse the membrane performance is. In particular, $R_2 = 1$ can be explained that the permeated hydrogen does not come from the WGS reaction at all but comes completely from the feed gas although in fact it is not. And the value of R_2 approximately equal to zero means the reactor is actually a traditional fixed-bed reactor. Fig. 6(b) shows the effect of *Da* and membrane permeability on hydrogen recovery R_2. As exhibited in the figure, R_2 increases with the increase of *Pp*, and almost reaches the maximum

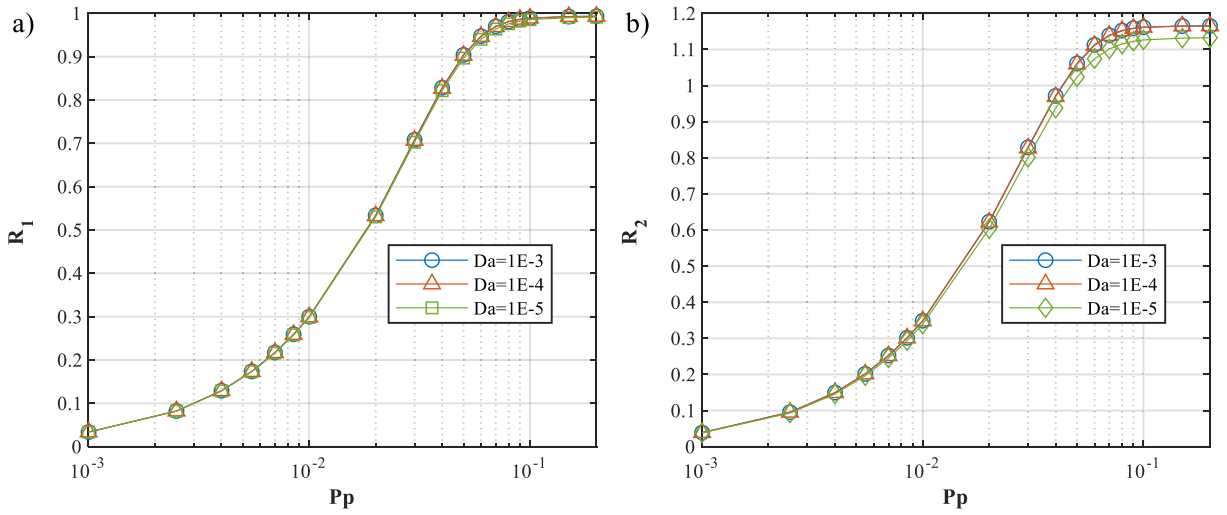


Fig. 6 – Effect of *Da* and *Pp* on the hydrogen recovery at T = 573 K.

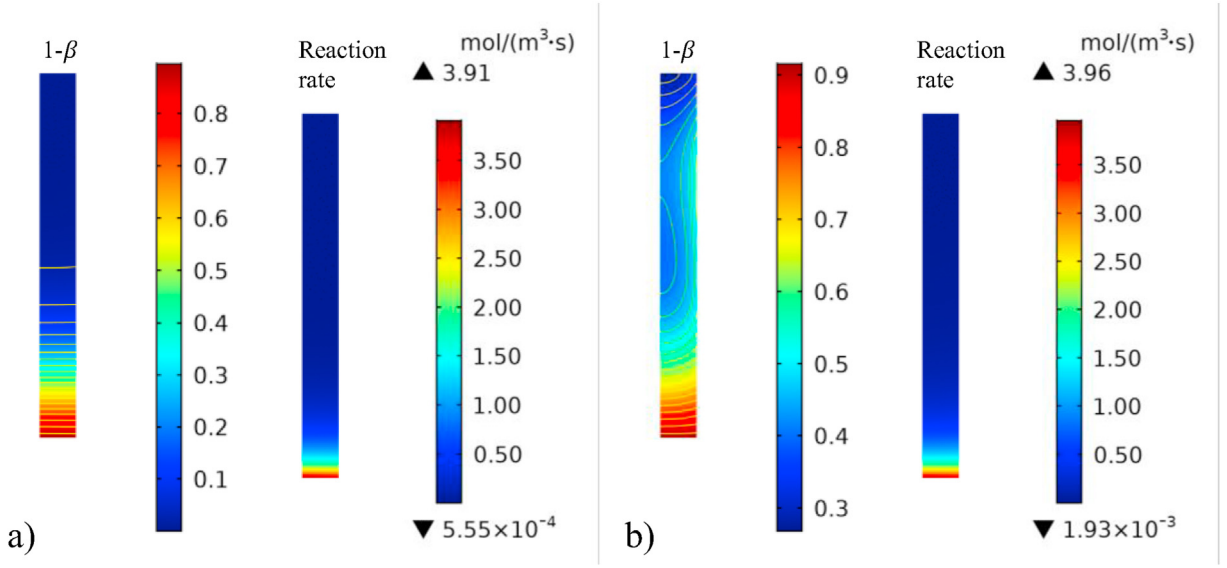


Fig. 7 – Distribution of $1-\beta$ and reaction rate at $Da = 10^{-4}$, T = 573 K, a) $Pp = 10^{-3}$, b) $Pp = 10^{-1}$.

value of 1.17 when $Pp = 10^{-1}$. Besides, R_2 is the same at $Da = 10^{-4}$ and $Da = 10^{-3}$, but obviously higher than that at $Da = 10^{-5}$ especially for $Pp \geq 10^{-1}$. This results indicate that the optimum values of *Pp* and *Da* are respectively 10^{-1} and 10^{-4} with regard to hydrogen recovery, and this set of values exactly yields the optimum CO conversion as observed in Figs. 3 and 4.

Reaction and hydrogen permeation phenomena in the reactor

Next, to gain some insight into the results above, we analyze how the overall performance of the reactor is determined by the reaction fields under conditions of different *Da* values and membrane permeability.

The chemical reaction rate is first examined because of different membrane permeability leading to variations of the WGS reaction equilibrium. Fig. 7 clearly shows the chemical equilibrium $1-\beta$ and reaction rate profile inside the membrane reactor. In contrast to the case of $Pp = 10^{-3}$ in Fig. 7(a), a notable increase of $1-\beta$ in Fig. 7(b) in the case of $Pp = 10^{-1}$ is observed especially in the area next to the membrane, indicating that the removal of hydrogen has an essential effect on the chemical equilibrium as expected. Unfortunately, a similar trend is not found in the reaction rate profiles. The reaction rate profiles are almost the same under different permeability as exhibited in Fig. 7(a)(b), except that the rate increases slightly under high permeability. This is because although removing hydrogen makes the WGS reaction far from equilibrium, the mole fractions of other components will inevitably change accordingly. That is, the reaction deviating far from the chemical equilibrium resulted from hydrogen separation may not necessarily lead to obvious increment of reaction rate in the area next to membrane.

Figure 8 shows the reaction rate distribution along the reactor (along reactor centerline). Based on the previous

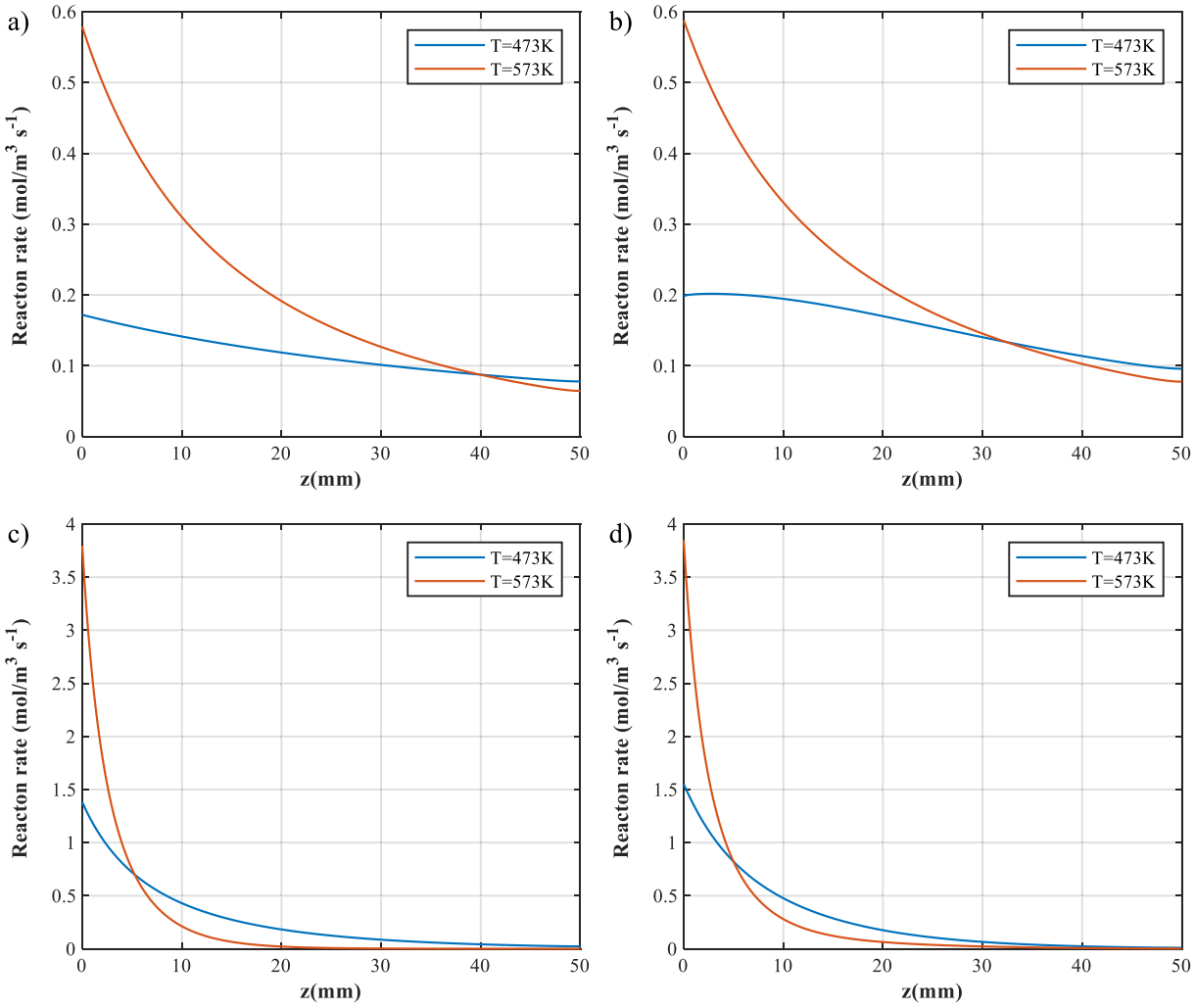


Fig. 8 – Reaction rate axially along the reactor at the conditions (a) $Da = 10^{-5}$, $Pp = 10^{-3}$ (b) $Da = 10^{-5}$, $Pp = 10^{-1}$ (c) $Da = 10^{-4}$, $Pp = 10^{-3}$ (d) $Da = 10^{-4}$, $Pp = 10^{-1}$.

discussion, in this analysis we fix the values of *Da* as 10^{-5} and 10^{-4} and *Pp* as 10^{-3} and 10^{-1}. As displayed in Fig. 8(a)(b), for $Da = 10^{-5}$ the reaction rate of T = 573 K is higher than that of T = 473 K in most of the reaction area, while in Fig. 8(c)(d), for $Da = 10^{-4}$ this phenomenon only occurs in the reactor inlet area accounting for about one-tenth of the reactor. This explains well the transition in Fig. 3(a) that at low *Pp* values Xco of high temperature is from greater than to lower than that of low temperature as *Da* increases from 10^{-5} to 10^{-4}. In contrast, in Fig. 3(b) Xco of high temperature is always greater than that of low temperature until it approaches 100% as *Da* increases, because the reaction is not limited by the thermodynamic equilibrium when $Pp = 10^{-1}$. Furthermore, from Fig. 8 (c)(d) it can be found that for $Da = 10^{-4}$ and T = 573 K the reaction rate is very high at the beginning and it decreases rapidly, dropping to almost zero in the middle of the reactor. This result shows that high temperatures can accelerate the WGS reaction and make it reach equilibrium more quickly, which is beneficial to reactor efficiency.

In fact, the trend of reaction rate distribution in Fig. 8(a)(b)(c)(d) is consistent, which reflects the characteristic of the WGS reaction kinetics. That is, the reaction rate of high temperature is distinctly greater than that of low temperature in the initial stage of reaction, but with the reaction proceeding, the reaction rate of high temperature drops rapidly and finally is lower than that of low temperature at the end of reaction. The difference is that for large *Da* values in the front of the reactor appears the reaction region where the reaction rate of high temperature and low temperature are equal, while as *Da* decreases this region will move backwards in the reactor as demonstrated in the figure.

Then, we focus on the mole fraction distribution of components. The hydrogen distribution in the entire reactor (retentate side and permeate side) is shown in Fig. 9. The profile of hydrogen distribution is relatively similar in the permeate side. Hydrogen enters the permeate side through the membrane and its content continuously increases along the sweep gas flow direction. In contrast, hydrogen distribution profiles in the retentate side are quite different at different *Da* values and membrane permeability. The extremely low mole fraction of hydrogen on the permeate side (order $10^{-4} - 10^{-2}$ with respect to the hydrogen mole fraction on the retentate side) indicates that the partial pressure of hydrogen can be neglected when high speed sweep gas is

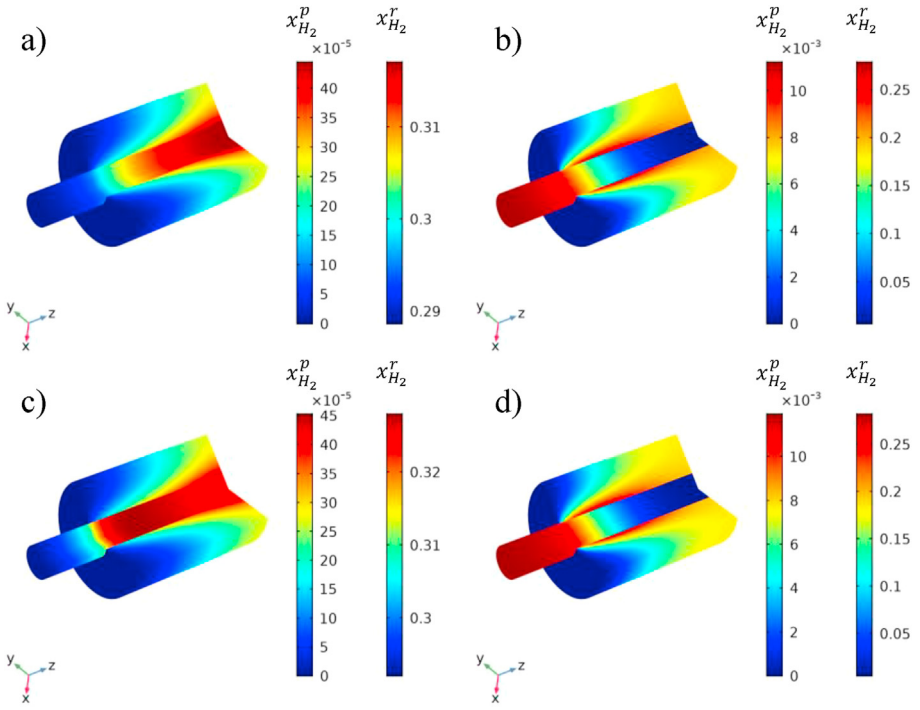


Fig. 9 – Profile of hydrogen mole fraction in the reactor at T = 573 K, (a) $Da = 10^{-5}$, $Pp = 10^{-3}$ (b) $Da = 10^{-5}$, $Pp = 10^{-1}$ (c) $Da = 10^{-4}$, $Pp = 10^{-3}$ (d) $Da = 10^{-4}$, $Pp = 10^{-1}$.

employed. So the flux of hydrogen permeating through the membrane will remain almost unaffected, whether co-current or counter-current sweep gas mode is configured [36].

Figure 10 shows the CO and H_2 molar fraction distribution of the whole reactor in the axial direction (along reactor centerline) at temperature of 573 K. It can be found that the molar fractions of components have already been different before entering the reaction zone, which is the result of diffusion caused by different concentration gradients due to different reaction conditions. Similar phenomenon is also reported in the literature [33]. In Fig. 10(a) for both the cases $Da = 10^{-4}$ and $Da = 10^{-3}$, the CO fraction curves have a similar shape and CO is completely consumed at the outlet of the reactor. In contrast, for $Da = 10^{-5}$, there is still almost 28%CO remaining in the reactor at the end of the reactor. This again demonstrates that $Da = 10^{-4}$ is an appropriate reaction condition for yielding high CO conversion under the configuration of this reactor, taking into account catalyst content and requirements of flow rates. Furthermore, the axial hydrogen content can be used as an indicator to evaluate the extent of the reaction. The position where the content of CO in the axial direction is close to zero indicates that the WGS reaction is almost complete here. Therefore, on the basis of the CO content at the outlet of the reactor, the proper geometric length of the reactor can be determined from the perspective of reaction conversion.

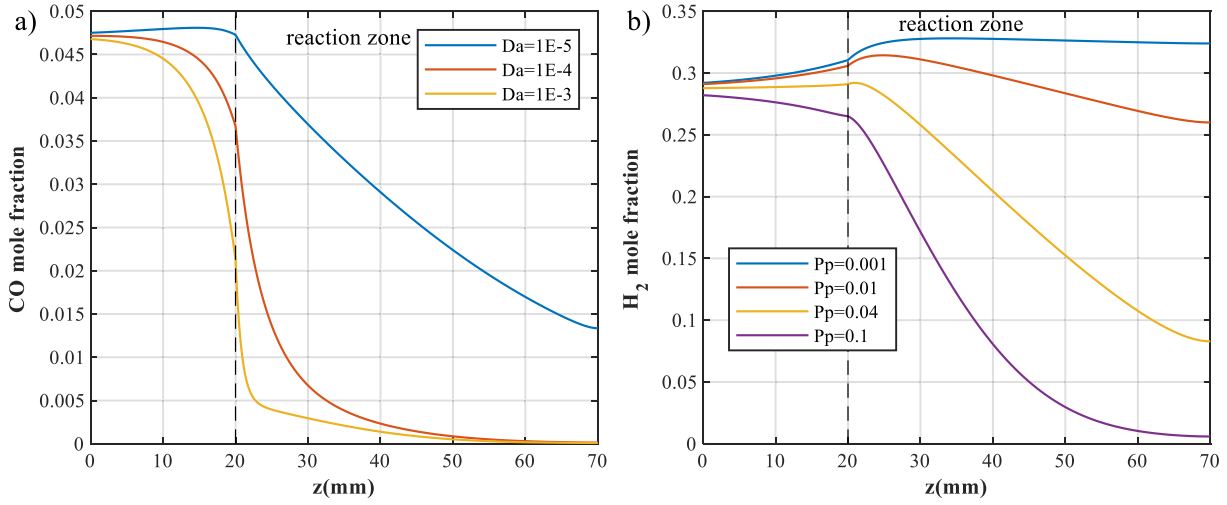


Fig. 10 – Axial distribution of (a) CO mole fraction at T = 573 K, $Pp = 10^{-1}$ (b) H_2 mole fraction at T = 573 K, $Da = 10^{-4}$.

Figure 10(b) shows the hydrogen molar fraction under different permeability. The mole fraction of hydrogen at the same position of the reactor decreases gradually with the increase of membrane permeability, as expected. From the figure, for $Pp = 10^{-3}$ along the axial direction the hydrogen mole fraction increases as the reaction proceeds and then gradually stabilizes as the reaction reaches equilibrium. As the membrane permeability continues to increase, for example, when $Pp = 10^{-2}$ the hydrogen content increases at the beginning and then decreases because the hydrogen separation rate begins to be greater than the hydrogen production rate of WGS reaction as the reaction gradually approaches equilibrium. When $Pp = 10^{-1}$, the hydrogen content decreases continuously from the beginning, and almost drops to zero at the reaction outlet indicating the separation of hydrogen is almost complete. Similarly, the axial hydrogen content at the outlet of the reactor can count as the indicator of the degree of hydrogen separation. As a supplement to the indicator of extent of WGS reaction, it can also provide guidance that cannot be ignored for the design of membrane reactor length.

Finally, we analyze the mole fraction of hydrogen along the radial direction at a certain axial position Z to investigate the mass transfer phenomenon outside the membrane. As shown in Fig. 11, the hydrogen mole fraction next to the membrane is about 90% or more of that in the center of the reactor, so it can be considered that the chemicals are evenly distributed in the radial direction. This result suggests that the diffusion of hydrogen inside the membrane reactor at low inlet velocity will not become the mass transfer resistance of hydrogen from the reaction zone to the permeation zone.

Effect of sweep gas flow on reactor performance

In the above, the effect of membrane permeability on reactor performance is investigated under ideal conditions utilizing the high speed sweep gas where the amount of hydrogen permeating through the membrane is not affected by the extremely low hydrogen content on the permeate side. Therefore, the effect of sweep gas (N_2) on the reactor performance is a point worth of attention, which is presented in Fig. 12. At membrane permeability $Pp = 10^{-1}$, for flow rate < 0.2L/min, hydrogen recovery R_2 increases rapidly with the increase of flow rates, while it gradually approaches to be constant for flow rate > 0.5L/min. This observation is also described by Boutikos et al. [35], indicating that there is an optimal range of flow rates of sweep gas, which should be about 0.2–0.5 L/min in this case taking into account hydrogen purity on the permeate side and reactor performance. However, with the decrease of membrane permeability the influence of sweep gas on R_2 weakens. As shown in the figure, when $Pp = 10^{-2}$ the sweep gas flow rate of 0.2 L/min result in only slight improvement of R_2 of 0.15, much lower than that of 0.65 when $Pp = 10^{-1}$. This shows that it is necessary to using the sweep gas for membrane reactors with large hydrogen permeation flux.

Next, the effect of co-current vs. counter-current sweep gas mode on the reactor performance is examined, as presented in Fig. 13 in the case of $Pp = 10^{-1}$ and $Da = 10^{-4}$. From the figure for flow rate < 0.01L/min, there is almost no performance difference between the two sweep gas modes, probably because the mass transfer of hydrogen on the permeate side is almost dominated by diffusion instead of convection on account of too low sweep gas speed. Also, the reactor performance is almost not affected by the sweep gas modes at high flow rates such as N2 flow rate = 1L/min. In contrast, the range of sweep gas flow rate is found to be 0.05–0.5 L/min in which the membrane reactor equipped with the counter current sweep mode has a pronounced performance lead over that with the co-current mode. Besides, it is worth noting that the differences in CO conversion is relatively small between two sweep gas modes as shown in Fig. 13(b), probably due to the high H_2O/CO ratio in the feed gas.

Figure 14 shows the driving force of hydrogen along the reactant flowing direction on both sides of the permeable membrane under two sweep gas modes. Due to the permeation effect the hydrogen partial pressure on the retentate side exhibits a gradual decrease, while on the permeate side it

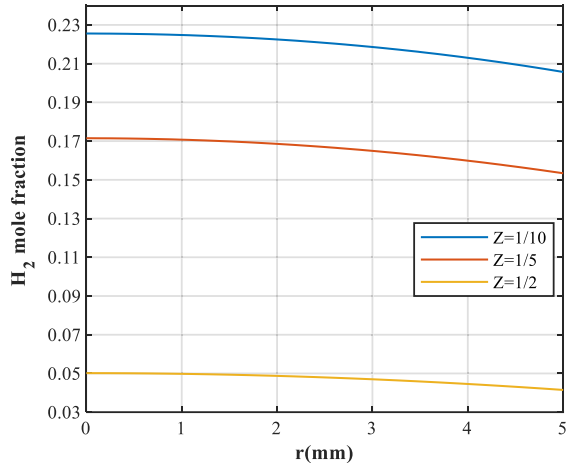


Fig. 11 – Hydrogen mole fraction along radical direction in the retentate side at $Da = 10^{-4}$, $Pp = 10^{-1}$, T = 473 K. Z = 1/10, 1/5, 1/2 respectively symbol one tenth, one fifth, one half of the reactor length.

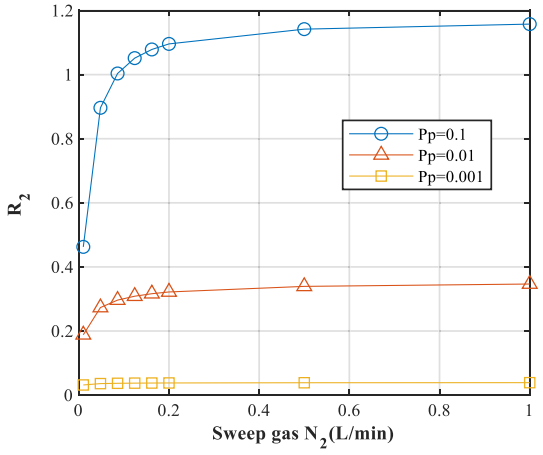


Fig. 12 – Effect of sweep gas flow rates on reactor performance R_2 at $Da = 10^{-4}$, T = 573 K.

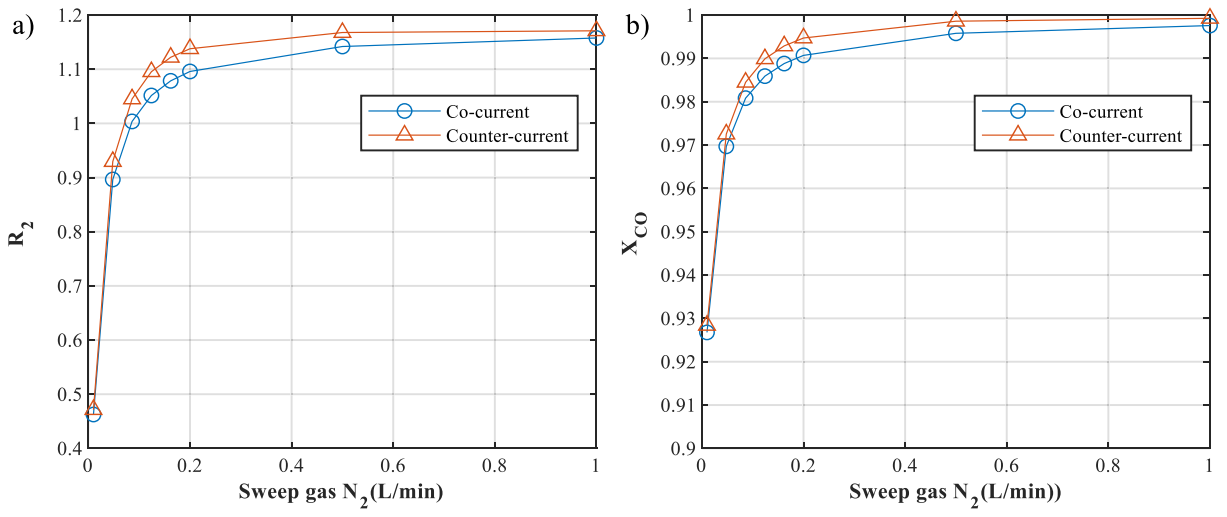


Fig. 13 – Effect of co-current vs. counter-current sweep gas mode on reactor performance at $Da = 10^{-4}$, $Pp = 10^{-1}$, T = 573 K.

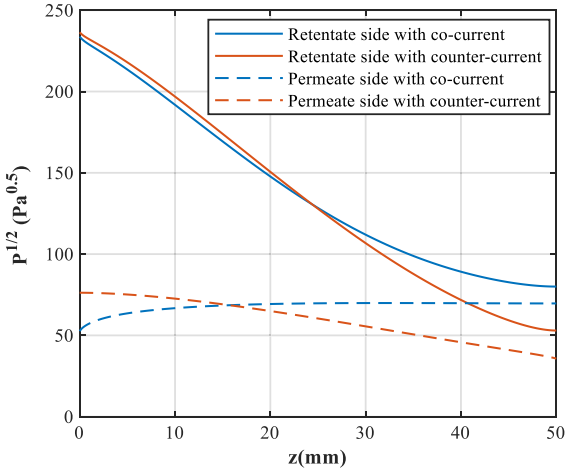


Fig. 14 – Driving force for hydrogen permeation on both retentate and permeate side at $Da = 10^{-4}$, $Pp = 10^{-1}$, T = 573 K, N_2 flow rate = 0.2L/min.

shows an opposite trend between the two sweep gas modes. In the co-current mode, the hydrogen partial pressure on the permeate side shows an upward trend, while in the counter-current mode, the hydrogen partial pressure on the permeate side continuously decreases as shown in the figure. This means that, compared with the co-current mode, the hydrogen partial pressure on the retentate side is allowed to drop to a lower level since the hydrogen partial pressure on the permeate side continues to decrease in the counter-current mode. In other words, assuming that the same pressure difference is maintained on both sides of the membrane, the hydrogen content at the outlet of the retentate side in the counter-current mode will be lower than in the co-current mode, and thus the improvement of the membrane reactor performance can be achieved.

Conclusion

The overall performance of the low-temperature water-gas reaction membrane reactor is studied by developing a CFD model based on the assumptions of isothermal reaction and laminar flow. To gain a better understanding of this issue, the reaction phenomenon inside the reactor is discussed in detail afterwards in terms of WGS reaction kinetics, Damkohler number (*Da*) and membrane permeability (*Pp*). We found that high operating temperatures are not only necessary to demonstrate the separation effect of the membrane from the perspective of thermodynamic equilibrium, but also can accelerate the reaction. At temperature of 573 K, the best *Da* and *Pp* values are found to be 10^{-4} and 10^{-1} respectively, corresponding to the optimal hydrogen recovery and CO conversion (nearly 100%). Then, we propose that taking the CO content and hydrogen content at the outlet of the reactor as two indicators of the degree of completion of the reaction and hydrogen separation, the optimal reactor design length can be well determined. Furthermore, the effect of sweep flow rates and sweep gas modes on the performance of the reactor is discussed. The results show that using sweep gas on the permeate side is necessary to improve the performance of the reactor with large hydrogen permeation flux. The most pronounced enhancement in reactor performance is observed for sweep gas flow rate of 0.2–0.5 L/min. And at this flow rate, further increase of reactor performance can be achieved by configuring the counter-current sweep gas mode. Finally, the research strategy in this article can also provide a reference for the research of different membrane reactors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

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Nomenclature

D_0	Diffusion coefficient, m^2/s
$D_{i,k}$	Multicomponent Fick diffusion coefficient, m^2/s
d	Reactor diameter, m
E_H	Activation energy, J/mol
F_{H_2}	Hydrogen flux through the membrane, mol/ $(m^2 \cdot s)$
k	Reaction rate constant, mol/(kg•h)
K_e	Equilibrium constant
M_0	Average molar mass, kg/mol
M_i	Molecular weight of component i, kg/mol
p	Pressure, Pa
P_0	Outlet pressure, Pa
P_i	Partial pressure of component i, Pa
p_1	Hydrogen partial pressure on the retentate side, Pa
p_2	Hydrogen partial pressure on the permeate side, Pa
P_L	Hydrogen permeability constant, $kg/(m \cdot s \cdot Pa^{0.5})$
P_m	Hydrogen permeance, $kg/(m^2 \cdot s \cdot Pa^{0.5})$
R	Gas constant, J/(mol•K)
R_i	Reaction rate of component i, $mol/(m^2 \cdot s)$
R_1	Ratio of the total permeated hydrogen to the total hydrogen
R_2	Ratio of the total permeated hydrogen to the hydrogen flowing into the reactor
r	Reaction rate of water-gas shift reaction, $mol/(m^3 \cdot s)$
T	Temperature, K
T_0	Inlet temperature, K
$\boldsymbol{u}$	Velocity vector, m/s
u_0	Inlet velocity, m/s
ω_k	Mass fraction of component i
x_k	Mole fraction of component i
X_{CO}	Conversion of carbon monoxide

Greek Letters

ε	Porosity of the catalyst bed
κ	Packed bed permeability, m^2
δ	Membrane thick, m
μ	Fluid viscosity, Pa•s
ρ	Fluid Density, kg/m^3
ρ_0	Inlet gas mixture density, kg/m^3

Dimensionless parameters

Da	Damkholer number
Pp	Membrane permeability
Pe	Peclet number

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